Kathleen Medriano

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EDUCATION

PhD, Cognitive Sciences September 2022 – Present
University of California, Irvine
Dissertation methods for time series data in the cognitive sciences.

Master of Science, Statistics September 2022 – December 2024
University of California, Irvine
Developed advanced skills in statistical inference, modeling, and data analysis with focus on Bayesian methods and time series analysis.

Graduate Diploma, Mathematics August 2017 – July 2019
University of the Philippines, Diliman
Built expertise in formal mathematical reasoning and proof techniques.

Bachelor of Science, Psychology June 2012 – June 2016
University of Santo Tomas
Foundational knowledge of human cognition and behavior.

RESEARCH EXPERIENCE

PhD Research Student September 2022 – Present
University of California, Irvine - Cognitive Sciences Department
Supervisor: Joachim Vandekerckhove
Developing Bayesian hierarchical models for intervention designs in mobile health applications. Creating statistical tests for diffusion processes and data-driven algorithms for model discovery in cognitive science.

Research Intern (CacTüS Program) July 2022 – September 2022
Max Planck Institute for Biological Cybernetics
Supervisor: Peter Dayan, Philip Schwartenbeck, Mihaly Banyai
Selected for the highly competitive CacTüS Internship program (acceptance rate: < 2%). Conducted research on generalization of learned structural knowledge across cognitive tasks using graph similarity in Dr. Peter Dayan's lab in Tübingen, Germany.

PUBLICATIONS

Manuscripts in Preparation / Under Review

Bayesian Ornstein-Uhlenbeck Models for Mobile Health Intervention Design In preparation
Kathleen Medriano, Joachim Vandekerckhove
Expected submission: 2025
Developing Bayesian hierarchical Ornstein-Uhlenbeck models for optimizing intervention timing and effectiveness in mobile health applications.

Peer-Reviewed Journal Articles

A Reconstruct-then-Bootstrap Test for the Sufficiency of Diffusion Processes 2026

Kathleen Medriano, Joachim Vandekerckhove
Computational Brain & Behavior
DOI: 10.1007/s42113-026-00268-w
A novel statistical test to assess whether a diffusion process model is sufficient for observed time series data, using Kramers-Moyal expansion and bootstrap methods.

The HDI+ ROPE decision rule is logically incoherent but we can fix it 2024

Alexander Etz, Andrés F. Chávez de la Peña, Luis Baroja, Kathleen Medriano, Joachim Vandekerckhove
Psychological Methods
Critical analysis and proposed solution for a commonly used Bayesian decision rule in psychological research.

Experiences of COVID-19-recovered healthcare workers in a tertiary hospital in the Philippines: a mixed-method inquiry 2022

MB Carascal, PE Capistrano, MDL Figueras, OLAC Cataylo, SMS Zuñiga, K Medriano
INQUIRY: The Journal of Health Care Organization, Provision, and Financing
Mixed-methods study examining the experiences and challenges faced by healthcare workers who recovered from COVID-19 in a Philippine tertiary hospital setting.

TALKS AND PRESENTATIONS

Model discovery for hierarchical datasets September 2026

Poster Presentation, International Conference on Scientific Computing and Machine Learning (SCML) 2026
Medriano, K., & Vandekerckhove, J. (2026, September). Model discovery for hierarchical datasets. Poster at SCML 2026.

Model discovery for hierarchical datasets July 2026

Conference Talk, MathPsych / ICCM 2026
Medriano, K., & Vandekerckhove, J. (2026, July). Model discovery for hierarchical datasets. Talk at MathPsych / ICCM 2026. https://mathpsych.org/presentation/2140

Reconstruct-then-Bootstrap test for the sufficiency of diffusion process June 2026

Poster Presentation, International Society for Bayesian Analysis 2026
Medriano, K., & Vandekerckhove, J. (2026, June). Reconstruct-then-Bootstrap test for the sufficiency of diffusion process. Poster at the International Society for Bayesian Analysis 2026.

Data-driven model discovery for coupled time series July 2025

Conference Talk, MathPsych / ICCM 2025
The Ohio State University, Ohio Union and Blackwell Inn
Introduced a data-driven framework for discovery of parsimonious stochastic differential equation models from coupled time series data. Demonstrated applications to core affect dynamics from daily life studies. 20-minute presentation (15 min + 5 min Q&A).

Amortized stochastic time series models for intervention designs November 2024

Poster Presentation, MathPsych @Psychonomics Satellite Meeting 2024

New York Marriott Marquis

Applied Ornstein-Uhlenbeck diffusion models to mobile health intervention data for psychological well-being in college students. Evaluated BayesFlow for simulation-based inference in hierarchical models.

Amortized stochastic time series models for intervention designs July 2024

Poster Presentation, MathPsych / ICCM 2024

Tilburg University, Cube building

Hierarchical Bayesian interrupted Ornstein-Uhlenbeck model applied to mobile health data. Examined intervention effectiveness across four study phases and explored BayesFlow implementation strategies.

Bayesian hierarchical Ornstein-Uhlenbeck models for intervention designs November 2023

Poster Presentation, SIDIC (Psychonomics-related meeting)

Grand Ballroom, Hilton San Francisco Union Square

Extended OU model analysis with BayesFlow implementation for simulation-based inference. Explored intervention effectiveness through Bayes factors on linear contrasts, revealing habituation patterns in attractor and elasticity parameters. Investigated BayesFlow applicability for covariates, multidimensional variates, and response artifacts.

Bayesian hierarchical Ornstein-Uhlenbeck models for intervention designs July 2023

Poster Presentation, MathPsych/ICCM/EMPG 2023

University of Amsterdam, Roeterseiland campus

Applied hierarchical Bayesian OU models to mobile health intervention data for psychological well-being. Demonstrated estimation of intervention effectiveness, persistence effects, and individual-level response features.

HONORS AND AWARDS

Women of Math Psych Travel Grant July 2026

Society for Mathematical Psychology

ISBA Junior Researcher Travel Award ($1,000) April 2026

International Society for Bayesian Analysis (ISBA), funded by NSF

Luce Family Fellowship in Mathematical and Behavioral Sciences ($10,000) April 2026

R. Duncan Luce Endowment

Tarow and Minako Indow Fellowship 2025

University of California, Irvine - Department of Cognitive Sciences

$7,000 fellowship awarded annually to two graduate students in recognition of exceptional research skills and productivity, as demonstrated during advancement to PhD candidacy. Awarded for research on 'Methods for Time Series Data in Cognitive and Psychological Sciences' under advisor Joachim Vandekerckhove.

Best Poster Award July 2024

MathPsych / ICCM 2024

Awarded for poster 'Amortized stochastic time series models for intervention designs' by Medriano, K., Oravecz, Z., & Vandekerckhove, J. Recognition for outstanding research presentation on Bayesian hierarchical models and BayesFlow implementation.

Best Poster Award July 2023

MathPsych/ICCM/EMPG 2023

Awarded for poster 'Bayesian hierarchical Ornstein-Uhlenbeck models for intervention designs' by Medriano, K., Oravecz, Z., & Vandekerckhove, J. Recognition for exceptional research on mobile health intervention modeling.

Magna Cum Laude Jun 2016

College of Science, University of Santo Tomas, Philippines

Academic distinction for graduating with honor from undergraduate studies.

Academic Excellence Award in Psychology June 2016

College of Science, University of Santo Tomas, Philippines

Awarded for graduating as the top student in the Psychology academic program.

Academic Program Award June 2016

UST Psychological Alumni Association, Philippines

Awarded for exhibiting exemplary academic performance throughout undergraduate studies.

Santo Tomas Academic Award Scholarship June 2012 – June 2016

University of Santo Tomas, Philippines

Full scholarship award for academic excellence; 100% tuition waived throughout entire undergraduate study (4 years).

PROFESSIONAL EXPERIENCE

Graduate Research Assistant September 2022 – Present

University of California, Irvine, Irvine, CA

- Conduct independent research on Bayesian hierarchical models and time series analysis
- Develop novel statistical methods for cognitive science applications
- Collaborate on mobile health intervention design projects
- Present research findings at academic conferences

Independent Researcher April 2021 – February 2022

Research Collaborations, Manila, Philippines

- Conducted statistical analyses (ANOVA, t-tests, etc.) using R for a study on Filipino attitudes towards LGBTQ+ individuals, focusing on homonegativity and sexual prejudice
- Collaborated with researchers from The Medical City (a tertiary hospital) to investigate the experiences of COVID-19-recovered healthcare workers
- Performed statistical analyses to assess the impact on healthcare worker well-being and work performance

Benchmark and Portfolio Database Associate January 2021 – April 2021

MSCI, Manila, Philippines

- Investigated and resolved data quality issues, identifying root causes (e.g., input errors, methodology issues)
- Improved data accuracy using DML and TCL
- Provided client support, addressing inquiries on data accuracy, methodology, and index computations

Benchmark and Portfolio Database Analyst April 2019 – January 2021

MSCI, Manila, Philippines

• Optimized Perl and Python codes for more efficient data processing and improved readability

• Developed an automated data processing script integrated with the Jira API

• Led and mentored the Philippine team, organizing meetings, overseeing projects, and training new hires

• Spearheaded a key project to adapt financial data pipelines for a new industry classification standard

• Coordinated code development, testing, deployment, and client communication to ensure seamless implementation

SKILLS

Programming Languages: R, Python, MATLAB, Stan, JAGS

Statistical Methods: Bayesian Methods, Time Series Analysis, Hierarchical Modeling, Machine Learning, Statistical Inference, Diffusion Processes

Software & Tools: RStudio, Jupyter, Git, LaTeX, Stan, BUGS/JAGS, SPSS, Markdown

Research Areas: Cognitive Sciences, Computational Modeling, Mobile Health, Model Discovery, Ornstein-Uhlenbeck Processes, Intervention Design

Languages: English (Native)